

Phase 1: Business Foundation

Healthcare Digital Platform — Business Analysis & Solution Design

Subtitle

Provider Lifecycle, Patient Care & Digital Healthcare Operations

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Role:

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Domain:

HealthTech | Provider Lifecycle | Patient Care | Digital Healthcare Operations

Document Version:

Version 2.0

Date:

June 2026 (update when finalized)

Document Control

Version	Date	Author	Description
1.0	June 2026	Vaibhav Yadav	Initial Healthcare Case Study
2.0	June 2026	Vaibhav Yadav	Enhanced Senior BA Documentation with Functional Artifacts

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1. Executive Summary

Project Background

This case study presents the business analysis and solution design of a digital healthcare platform covering provider onboarding, verification, activation, and downstream healthcare operations.

The platform addresses challenges such as manual provider verification, multiple operational handoffs, limited process visibility, and delays across interconnected provider and patient processes. The proposed solution standardizes these processes through digital workflows, automated verification and business rules, exception handling, integrated services, notifications, SLA monitoring, and operational reporting.

Once a provider is verified and activated, the platform supports downstream activities including appointment management, patient consultation, and payment processing. Operational teams can monitor application status, SLA performance, and key business metrics through centralized dashboards.

The case study covers requirements analysis, process modelling, functional requirements, business rules, solution architecture, data modelling, API specifications, Agile user stories, acceptance criteria, UAT, SQL analytics, dashboards, and governance.

2. Business Problem

The existing digital healthcare platform relies on multiple manual and fragmented processes across provider onboarding, verification, appointment management, consultation, payment, notifications, and operational support. Provider credential verification, document validation, compliance processing, and approval workflows require multiple handoffs, while downstream healthcare processes depend on successful provider activation. This can result in operational bottlenecks, delayed provider activation, inconsistent service levels, fragmented workflows, and limited visibility across the healthcare lifecycle.

Key challenges include:

- Manual verification of provider credentials.
- Long onboarding turnaround times.
- Limited visibility into pending verification cases.
- High operational effort for compliance teams.
- Inconsistent SLA adherence.
- Delayed provider activation impacts business growth.
- Lack of centralized reporting for operational performance.
- Fragmented appointment and consultation workflows.
- Payment and notification dependencies across the healthcare journey.

- Limited end-to-end operational visibility.

These inefficiencies reduce operational productivity, affect provider and patient experience, and limit the platform's ability to scale its digital healthcare operations.

3. Business Objectives

The project aims to modernize the provider onboarding journey through workflow automation and standardized operational processes.

Primary Objectives

- Digitize and standardize provider registration and onboarding workflows.
- Automate credential and regulatory verification wherever applicable.
- Improve provider approval, compliance processing and activation turnaround time.
- Enable standardized appointment and consultation workflows after provider activation.
- Support integrated payment and notification workflows across the healthcare journey.
- Provide real-time operational visibility through dashboards, SLA monitoring and KPI reporting.
- Improve auditability, access control and compliance tracking across critical workflows.
- Reduce manual operational effort and improve the overall provider and patient experience.

4. Project Scope

In Scope

Provider Lifecycle

- Provider registration and profile creation
- Credential and document submission/validation
- Regulatory verification
- Automated approval/rejection and compliance exception handling
- Provider activation

Patient & Healthcare Operations

- Patient management
- Appointment management
- Consultation management
- Prescription/consultation completion workflow

- Payment processing and refund workflow
- Notifications
- Support ticket handling

Operations & Governance

- SLA monitoring
- Operational dashboards and KPI reporting
- Audit logs
- Role-based access
- Compliance tracking

Out of Scope

- Electronic Health Records (EHR)
- Pharmacy management
- Laboratory management
- Hospital administration
- Claims processing
- Payroll management
- Insurance reimbursement workflows

Phase 2 – Business Analysis & Process Discovery

5. Stakeholder Analysis

Stakeholder Matrix

Stakeholder	Role	Key Responsibilities	Primary Expectations
Patients	End Users	Book appointments, consult providers, make payments	Fast access to verified providers and seamless booking experience
Providers	Service Providers	Register, complete onboarding, deliver consultations	Faster onboarding and timely activation
Operations Team	Operations	Verify documents, manage onboarding queue, resolve exceptions	Reduced manual effort and improved SLA compliance

Compliance Team	Regulatory Oversight	Validate licenses, ensure regulatory compliance, conduct audits	Accurate verification and complete audit trail
Support Team	Customer Support	Resolve provider and patient issues, manage escalations	Faster issue resolution and improved customer satisfaction
Product Team	Product Management	Define business requirements, prioritize enhancements	Improved platform adoption and business outcomes
Engineering Team	Development	Build integrations, workflows and platform features	Clear functional requirements and stable integrations

Stakeholder RACI Matrix

Activity	Product	Operations	Compliance	Engineering	Support
Requirement Gathering	A	C	C	C	I
Provider Verification	I	R	A	I	I
API Integration	C	I	C	R/A	I
Dashboard Development	A	C	I	R	I
UAT	A	R	C	C	R
Production Deployment	C	I	I	A/R	C

Legend:

- R – Responsible
- A – Accountable
- C – Consulted
- I – Informed

6. Current Process (AS-IS)

Current Workflow

The existing provider onboarding process relies heavily on manual activities across Operations and Compliance teams. Each onboarding request passes through multiple validation stages before a provider becomes active on the platform.

Current Process

- The provider submits registration details.
- Supporting documents are uploaded.
- Operations perform initial validation.
- Compliance manually verifies medical registration.
- Missing information is communicated manually.
- The provider submits corrections.
- Operations complete review.
- Provider profile is activated.

Current Challenges

Challenge	Business Impact
Manual verification	Longer onboarding time
Multiple handoffs	Operational delays
No automated validation	Higher error rate
Lack of real-time tracking	Poor visibility
Manual communication	Increased operational effort
Limited dashboards	Difficult performance monitoring
SLA breaches	Reduced provider satisfaction

Reference Diagram

Refer to Asset 1 – Slide 16 (AS-IS Swimlane Process Diagram).

7. Future Process (TO-BE)

Proposed Workflow

The proposed onboarding solution introduces workflow automation, API-driven verification, business rules, and operational dashboards to reduce manual intervention while improving compliance.

Proposed Process

1. Provider registers.
2. Documents are uploaded.
3. System validates mandatory fields.
4. NMC Verification API triggered.
5. Automatic verification response received.
6. Business Rules Engine evaluates outcomes.
7. Low-risk providers auto-approved.
8. Exceptions routed to Compliance.
9. Notifications are sent automatically.
10. Provider profile is activated.
11. Dashboard updated in real time.

Expected Improvements

Current State	Future State
Manual verification	Automated verification
Manual routing	Rule-based routing
Manual communication	Automated notifications
Static reporting	Live dashboards
Longer turnaround	Reduced onboarding time
Limited visibility	Real-time monitoring

Business Benefits

- Reduced onboarding turnaround time.
- Improved SLA compliance.
- Faster provider activation.
- Better regulatory compliance.
- Reduced operational workload.
- Improved provider experience.
- Enhanced reporting and decision-making.

Reference Diagram

Refer to Asset 1 – Slide 17 (TO-BE Swimlane Process Diagram).

Phase 3 – Requirements Engineering

8. Business Requirements

Business Requirement Summary

BR ID	Business Requirement	Business Objective
BR-01	Enable digital provider registration through a centralized onboarding portal.	Improve provider onboarding experience
BR-02	Automate provider for credential verification using external regulatory registries.	Reduce manual verification effort
BR-03	Standardize onboarding workflows with configurable business rules.	Improve process consistency
BR-04	Route verification exceptions to Compliance for manual review.	Ensure regulatory compliance
BR-05	Provide onboarding status updates to providers through automated notifications.	Enhance user experience
BR-06	Enable automated notifications via Email/SMS/WhatsApp.	Improve communication efficiency
BR-07	Capture operational metrics for SLA and KPI reporting.	Support business decision-making
BR-08	Maintain complete audit logs for all onboarding activities.	Ensure compliance and traceability

9. Functional Requirements

Module 1 – Provider Registration

FR ID	Functional Requirement
FR-01	The system shall allow providers to register using personal, professional, and contact information.
FR-02	The system shall validate all mandatory fields before submission.
FR-03	The system shall generate a unique Provider ID upon successful registration.

Module 2 – Provider Verification

FR ID	Functional Requirement
FR-04	The system shall automatically invoke the Provider Verification API upon profile submission.
FR-05	The system shall validate registration details against the National Medical Council for registry.
FR-06	The system shall automatically approve providers with an Active license status.
FR-07	The system shall route Suspended or Expired licenses to the Compliance queue for manual review.

Module 3 – Appointment Activation

FR ID	Functional Requirement
FR-08	The system shall activate provider appointment scheduling after successful onboarding approval.
FR-09	The system shall prevent appointment booking for inactive providers.

Module 4 – Notifications

FR ID	Functional Requirement
FR-10	The system shall send onboarding status notifications through Email, SMS, or WhatsApp.
FR-11	The system shall notify Operations for pending verification beyond SLA thresholds.

Module 5 – Dashboard & Reporting

FR ID	Functional Requirement
FR-12	The system shall display onboarding KPIs in an executive dashboard.
FR-13	The system shall track Approval Rate, Rejection Rate, Pending Queue, Activation Rate, and SLA Compliance.

Module 6 – Payment & Consultation

FR ID	Functional Requirement
FR-14	The system shall initiate payment processing for booked consultations.
FR-15	The system shall update payment status based on gateway response.
FR-16	The system shall create consultation records after successful appointment completion.

10. Non-Functional Requirements

Category	Requirement
Performance	Verification responses should be processed within defined SLA limits where external services are available.

Availability	The platform should be available 24x7, excluding scheduled maintenance windows.
Scalability	The solution should support increasing provider onboarding volumes without significant performance degradation.
Reliability	Failed API requests should be retried and logged for operational review.
Security	Sensitive provider and patient data should be encrypted in transit and at rest.
Auditability	All onboarding lifecycle events should be recorded for audit purposes.
Usability	The onboarding workflow should be intuitive and minimize manual effort.
Maintainability	Business rules should be configurable with minimal development effort.

11. Business Rules

Rule ID	Business Rule
BR-Rule-01	A provider cannot be activated without successful regulatory verification.
BR-Rule-02	Providers with Suspended or Expired licenses must be routed to the Compliance queue.
BR-Rule-03	Appointment booking is permitted only by Active providers.
BR-Rule-04	Payment must be successfully completed before appointment confirmation.
BR-Rule-05	Pending verification cases exceeding the defined SLA must be escalated to Operations.
BR-Rule-06	Every onboarding status change must generate an audit log entry.

BR-Rule-07	Only authorized roles can approve or reject provider applications.
BR-Rule-08	Dashboard metrics must refresh based on the latest onboarding transaction data.

Phase 4 – Functional Design

12. Functional Architecture

System Overview

The Healthcare Digital Platform is designed using a layered architecture to separate user interactions, business processes, integrations, and data management. This approach improves maintainability, scalability, and integration with external healthcare services.

High-Level Functional Architecture

Layer	Components	Purpose
Presentation Layer	Provider Portal, Patient Portal, Operations Dashboard, Compliance Dashboard, Admin Portal	User interaction and workflow management
Application Layer	Provider Onboarding Module, Appointment Management Module, Consultation Module, Payment Module, Notification Module, Support Module	Executes business processes and workflow logic
Integration Layer	Provider Verification API (NMC), Payment Gateway API, Notification API (Email/SMS/WhatsApp)	Enables communication with external systems
Data Layer	Providers, Patients, Appointments, Consultations, Payments, Support Tickets, Onboarding Transactions	Stores operational and transactional data

Functional Flow

1. Provider registers through the onboarding portal.
2. System validates mandatory information.
3. Provider Verification API validates registration details.
4. Business Rules Engine evaluates verification responses.
5. Provider is Approved, Rejected, or routed for Manual Review.
6. Notification service informs stakeholders.
7. Approved providers become available for appointments.
8. Payment and consultation workflows continue.
9. Dashboard updates operational KPIs.

13. Entity Relationship Model (ERD)

Database Overview

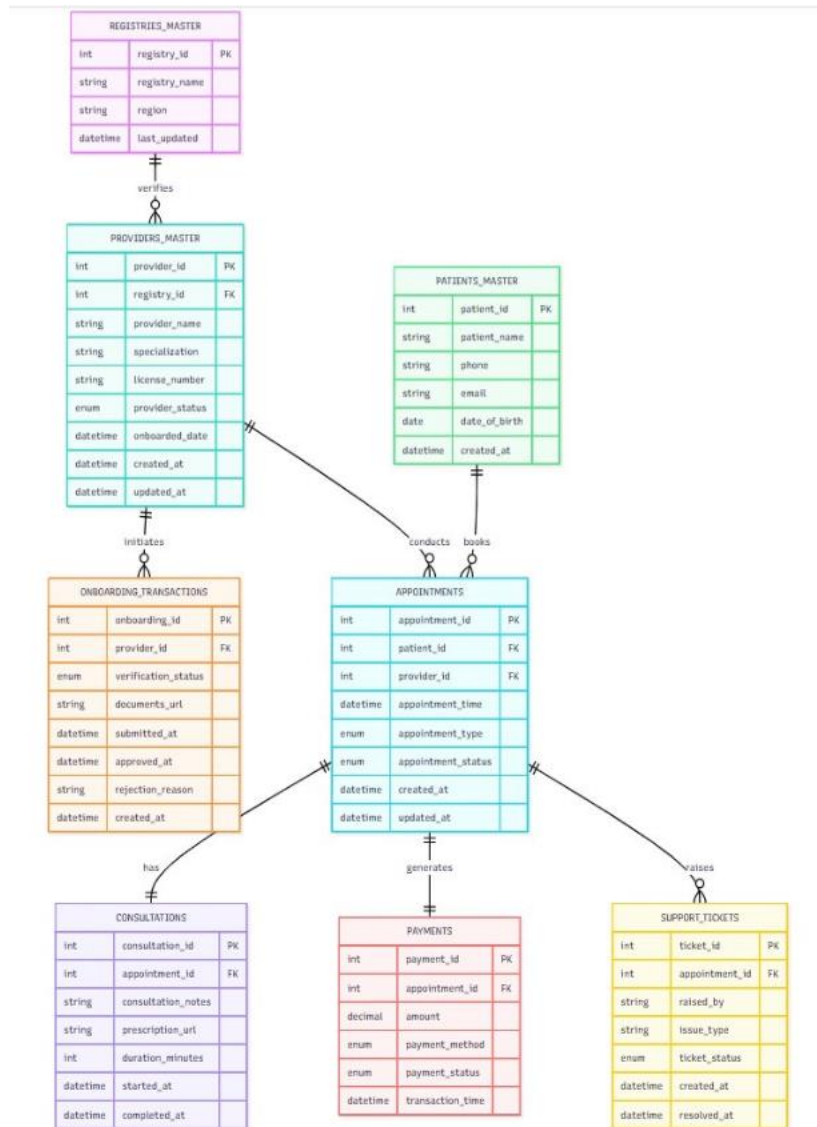
The platform uses eight primary entities to support onboarding, appointment management, consultation tracking, payments, customer support, and operational reporting.

Data Model Scope: The solution architecture defines an 8-entity conceptual data model covering the broader healthcare platform. The executable PostgreSQL schema focuses on the three core entities required for the provider onboarding and analytics use case: Registries Master, Providers Master, and Onboarding Transactions.

Relationship Summary

Parent Entity	Relationship	Child Entity
Registries Master	1 : Many	Providers Master
Providers Master	1 : Many	Onboarding Transactions
Patients Master	1 : Many	Appointments

ERD Reference



14. Data Dictionary

Table 1 – Registries Master

Field	Data Type	Description
registration_number	VARCHAR(30)	Unique medical registration number
doctor_name	VARCHAR(100)	Registered provider name
medical_specialty	VARCHAR(50)	Clinical specialization
license_status	VARCHAR(20)	Active, Suspended, Expired

Table 2 – Providers Master

Field	Data Type	Description
provider_id	INT	Primary Key
registration_number	VARCHAR(30)	Foreign Key
provider_name	VARCHAR(100)	Provider name
specialty	VARCHAR(50)	Medical specialty
network_status	VARCHAR(20)	In-Network / Out-of-Network
activation_status	VARCHAR(20)	Active / Inactive

Table 3 – Patients

Field	Data Type	Description
patient_id	INT	Primary Key
patient_name	VARCHAR(100)	Patient name
mobile_number	VARCHAR(15)	Contact number
email	VARCHAR(100)	Email address

Table 4 – Onboarding Transactions

Field	Data Type	Description
application_id	INT	Primary Key
provider_id	INT	Foreign Key
registration_number	VARCHAR(30)	Foreign Key
submission_timestamp	TIMESTAMP	Registration submitted
verification_timestamp	TIMESTAMP	Verification completed
approval_timestamp	TIMESTAMP	Approval completed
application_status	VARCHAR(20)	Approved / Rejected / Pending
rejection_reason	VARCHAR(100)	Reason for rejection

Table 5 – Appointments

Field	Data Type	Description
appointment_id	INT	Primary Key
patient_id	INT	Foreign Key
provider_id	INT	Foreign Key
booking_timestamp	TIMESTAMP	Appointment booking time
appointment_status	VARCHAR(20)	Scheduled / Completed / Cancelled

Table 6 – Consultations

Field	Data Type	Description
consultation_id	INT	Primary Key
appointment_id	INT	Foreign Key
consultation_date	TIMESTAMP	Consultation date
consultation_status	VARCHAR(20)	Completed / Cancelled
prescription_summary	TEXT	Consultation summary

Table 7 – Payments

Field	Data Type	Description
payment_id	INT	Primary Key
appointment_id	INT	Foreign Key
payment_amount	DECIMAL(10,2)	Consultation amount
payment_status	VARCHAR(20)	Success / Failed / Refunded
payment_method	VARCHAR(30)	UPI / Card / Net Banking

Table 8 – Support Tickets

Field	Data Type	Description
ticket_id	INT	Primary Key
appointment_id	INT	Foreign Key
issue_category	VARCHAR(50)	Payment Failure, SLA Breach, Technical Issue
ticket_status	VARCHAR(20)	Open / In Progress / Closed
created_timestamp	TIMESTAMP	Ticket creation time

Data Flow Summary

Source	Destination	Purpose
Provider Portal	Providers Master	Provider registration
Provider Verification API	Registries Master	License verification
Onboarding Module	Onboarding Transactions	Workflow tracking
Appointment Module	Appointments	Booking management
Consultation Module	Consultations	Consultation records
Payment Gateway	Payments	Payment tracking
Support Module	Support Tickets	Issue management
Analytics Engine	Dashboard	KPI reporting

Phase 5 – Agile Backlog & User Stories

15. Agile Product Backlog

Epic 1 – Provider Registration & Onboarding

Enable healthcare providers to complete a secure, digital onboarding process with minimal manual intervention.

US-HEALTH-01: Provider Registration

As a healthcare provider
I want to register using my professional and personal information
So that I can initiate the onboarding process.

Acceptance Criteria (Gherkin)

Given the provider is on the registration page
When all mandatory details are entered
Then the system shall create a provider profile
And generate a unique Provider ID.

US-HEALTH-02: Document Upload

As a healthcare provider
I want to upload mandatory verification documents
So that my profile can be verified.

Acceptance Criteria

Given the provider has completed registration
When required documents are uploaded
Then the application shall allow profile submission.

US-HEALTH-03: Profile Validation

As an Operations Executive
I want mandatory field validation before submission
So that incomplete applications do not enter the verification queue.

Acceptance Criteria

Given mandatory information is missing
When the provider clicks Submit
Then the application shall display validation errors
And prevent submission.

Epic 2 – Provider Verification & Compliance

Automate provider verification while ensuring regulatory compliance.

US-HEALTH-04: Provider Verification API

As an Operations Executive

I want provider credentials verified automatically

So that manual verification effort is reduced.

Acceptance Criteria

Given onboarding has been submitted

When the system invokes the NMC Verification API

Then the provider status shall be updated based on the API response.

US-HEALTH-05: Auto Approval

As a Compliance Manager

I want providers with valid licenses to be automatically approved

So that onboarding time is reduced.

Acceptance Criteria

Given the API returns an Active license

When verification completes

Then the application shall set the provider status to Approved

And activate the provider profile.

US-HEALTH-06: Compliance Queue

As a Compliance Executive

I want rejected or suspicious applications routed for manual review

So that regulatory risks are minimized.

Acceptance Criteria

Given verification fails

When the provider has an Expired or Suspended license

Then the application shall move the request to the Compliance Queue.

Epic 3 – Appointment & Consultation Management

Enable verified providers to manage appointments and complete consultations through a structured workflow.

US-HEALTH-07: Appointment Activation

As a verified healthcare provider

I want my appointment calendar to become available immediately after successful onboarding

So that patients can start booking consultations.

Acceptance Criteria

Given the provider's onboarding status is Approved

When the activation process completes

Then the provider profile shall be marked as Active

And appointment slots shall become available for booking.

US-HEALTH-08: Consultation Completion

As a healthcare provider

I want to mark consultations as completed

So that consultation records are maintained accurately.

Acceptance Criteria

Given an appointment is in Scheduled status

When the provider clicks Complete Consultation

Then the appointment status shall change to Completed

And a consultation record shall be created.

US-HEALTH-09: Prescription Upload

As a healthcare provider

I want to upload consultation notes and prescriptions

So that patients can securely access their treatment information.

Acceptance Criteria

Given a consultation is completed

When the provider uploads prescription details

Then the prescription shall be linked to the consultation record

And be accessible to the patient.

Epic 4 – Payments & Notifications

Enable secure payment processing and automated communication throughout the provider and patient journey.

US-HEALTH-10: Payment Processing

As a patient

I want to complete payment securely before my consultation

So that my appointment is confirmed.

Acceptance Criteria

Given a patient proceeds to checkout

When payment is successfully authorized

Then the payment status shall become Success

And the appointment status shall change to Scheduled.

US-HEALTH-11: Automated Notifications

As a provider and patient

I want to receive automated onboarding and appointment notifications

So that I remain informed about important status updates.

Acceptance Criteria

Given an onboarding or appointment status changes

When the event is processed

Then the system shall send Email, SMS, or WhatsApp notifications

And record the notification status.

US-HEALTH-12: Refund Processing

As a patient

I want to receive an automatic refund if my appointment is cancelled

So that I do not need to manually request reimbursement.

Acceptance Criteria

Given an appointment is cancelled

When payment has already been completed

Then the system shall initiate the refund process

And update the payment status to Refunded.

Epic 5 – Operations & Reporting

Provide operational visibility, SLA monitoring, and proactive issue management through dashboards and support workflows.

US-HEALTH-13: SLA Monitoring

As an Operations Manager

I want to monitor onboarding requests approaching SLA deadlines

So that delays can be addressed proactively.

Acceptance Criteria

Given an onboarding request exceeds the defined SLA threshold

When the dashboard refreshes

Then the application shall highlight the request

And notify the Operations team.

US-HEALTH-14: Executive Dashboard

As a Business Manager

I want to view operational KPIs through an executive dashboard

So that I can monitor business performance and make informed decisions.

Acceptance Criteria

Given dashboard data is refreshed

When the manager accesses the dashboard

Then KPIs including Approval Rate, Rejection Rate, Pending Queue, Activation Rate, and SLA

Compliance shall be displayed.

US-HEALTH-15: Support Ticket Escalation

As a Support Executive

I want failed onboarding, payment, or system issues to automatically generate support tickets So that customer issues are resolved efficiently.

Acceptance Criteria

Given a payment failure, API failure, or onboarding exception occurs
When the system detects the event
Then a support ticket shall be created automatically
And assigned to the appropriate support queue.

US-HEALTH-16: Audit Trail

As a Compliance Manager,
I want all onboarding status changes and actions to be recorded in an audit trail,
so that onboarding activity can be traced for compliance and review

Acceptance Criteria

Given an onboarding application exists
when the onboarding status changes
Then the system shall create an audit log entry for the status change
And the audit trail shall be available for authorized review

16. Agile Release Planning

Sprint	Features
Sprint 1	Registration, Login, Document Upload
Sprint 2	Verification API, Compliance Workflow
Sprint 3	Appointment & Consultation
Sprint 4	Payments & Notifications
Sprint 5	Dashboard, Reporting & UAT

Definition of Ready (DoR)

A user story is considered ready when:

- Business objective is defined.
- Acceptance criteria are documented.
- Dependencies are identified.
- UI/Process flow is available (if applicable).
- Stakeholder approval is obtained.

Definition of Done (DoD)

A user story is considered complete when:

- Functional requirements are implemented.
- Acceptance criteria are satisfied.
- UAT is successfully completed.
- No critical defects remain.
- Business stakeholder approves the feature.

Phase 6 – Integration Specifications

17. Integration Overview

Purpose

The Healthcare Digital Platform integrates with external services to automate provider verification, payment processing, and stakeholder notifications. These integrations reduce manual effort, improve processing speed, and ensure seamless communication between systems.

Integration	Purpose	Trigger	Response
Provider Verification API	Validate provider credentials with National Medical Council (NMC)	Provider submits onboarding application	Verification status (Approved / Rejected / Pending)
Payment Gateway API	Process consultation payments	Patient confirms appointment	Payment Success / Failure / Refunded
Notification API	Send Email/SMS/WhatsApp notifications	Status change or system event	Delivery confirmation

18. API Contract 1 – Provider Verification API

Business Purpose

Automatically validate provider registration details against the National Medical Council (NMC) registry before onboarding approval.

API Details

Attribute	Value
Method	POST
Endpoint	/provider/verify
Authentication	OAuth 2.0 Bearer Token
Content Type	application/json

Sample Request

```
{  
  "registration_number": "MH12345", "provider_name": "Dr. Priya Sharma",  
  "state_council": "Maharashtra Medical Council"  
}
```

Sample Success Response

```
{  
  "status": "SUCCESS", "license_status": "Active",  
  "verification_result": "Verified", "action": "AUTO_APPROVE"  
}
```

Sample Failure Response

```
{  
  "status": "FAILED",  
  "license_status": "Suspended",  
  "verification_result": "Not Verified",  
}
```

```
"action": "MANUAL_REVIEW",  
"reason": "License Suspended"  
}
```

Business Rules

- Only Active licenses qualify for automatic approval.
- Suspended or Expired licenses are routed to the Compliance queue.
- All verification attempts are logged for audit purposes

.

19. API Contract 2 – Payment Gateway API

Business Purpose

Enable secure payment processing for patient consultations.

API Details

Attribute	Value
Method	POST
Endpoint	/payments/process
Authentication	API Key
Content Type	application/json

Sample Request

```
{  
  
  "appointment_id": 5001,  
  "patient_id": 301,  
  "provider_id": 101,  
  "amount": 1500, "currency": "INR"  
}
```

Sample Success Response

```
{  
  "status": "SUCCESS",  
  "payment_reference": "PAY123456",  
  "payment_status": "Completed"  
}
```

Sample Failure Response

```
{  
  "status": "FAILED",  
  "payment_status": "Declined",  
  "reason": "Insufficient Balance"  
}
```

Business Rules

- Appointment confirmation occurs only after successful payment.
- Failed payments allow retry without duplicate booking creation.
- Refunds are initiated automatically for cancelled appointments (where applicable).

20. API Contract 3 – Notification API

Business Purpose

Notify providers and patients about onboarding, appointments, payments, and operational updates.

Attribute	Value
Method	POST
Endpoint	/notifications/send
Authentication	API Key
Channels	Email, SMS, WhatsApp

Sample Request

```
{  
  "recipient": "+919999999999",  
  "channel": "WhatsApp",  
  "template": "Provider_Approved",  
  "provider_name": "Dr. Priya Sharma"  
}
```

Sample Success Response

```
{  
  "status": "DELIVERED",  
  "message_id": "MSG458923"  
}
```

Sample Failure Response

```
{  
  "status": "FAILED",  
  "reason": "Invalid Mobile Number"  
}
```

Business Rules

- Notifications are triggered after onboarding, payment, and appointment events.
- Delivery failures are logged for operational review.
- Multiple channels (Email, SMS, WhatsApp) can be configured based on user preference.

Integration Sequence

Provider

|



Provider Portal

|



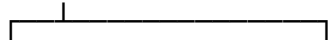
Verification API

|



Business Rules Engine

|



|

|

Approved Manual Review

|

|



Notification Compliance Queue

|



Provider Activated

Integration Error Handling

Scenario	System Action
Verification API timeout	Retry request; if unsuccessful, move application to Pending and notify Operations
Payment failure	Keep appointment unconfirmed and allow payment retry
Notification failure	Log the failure and retry through an alternate configured channel

Phase 7 – Testing & Validation

21. User Acceptance Testing (UAT)

UAT Objective

The objective of User Acceptance Testing (UAT) is to validate that the Healthcare Digital Platform meets business requirements and is ready for production deployment. UAT is performed by business users to ensure the solution supports operational workflows, regulatory requirements, and user expectations.

UAT Entry Criteria

- Functional development completed.
- System Integration Testing (SIT) completed successfully.
- Test environment available.
- Test data prepared.
- Business requirements approved.
- Test cases reviewed and signed off.

UAT Test Cases

UAT ID	Feature	Test Scenario	Expected Result	Status
UAT-01	Provider Registration	Register a new provider with valid details	Provider profile created successfully	Planned
UAT-02	Document Upload	Upload mandatory verification documents	Documents uploaded successfully	Planned
UAT-03	Provider Verification	Verify an Active provider via NMC API	Provider automatically approved	Planned
UAT-04	Compliance Queue	Verify a Suspended provider	Application routed to Compliance	Planned

UAT-05	Appointment Activation	Activate appointment after approval	Provider becomes available for booking	Planned
UAT-06	Payment Processing	Complete payment successfully	Payment status updated to Success	Planned
UAT-07	Notification Service	Send onboarding approval notification	WhatsApp/Email delivered	Planned
UAT-08	Dashboard	Verify KPI dashboard updates	Latest onboarding metrics displayed	Planned
UAT-09	SLA Monitoring	Delay onboarding beyond SLA	Escalation alert generated	Planned
UAT-10	Support Ticket	Trigger payment failure	Support ticket created automatically	Planned
UAT-11	Audit Trail	Verify that onboarding status changes and user actions are recorded	All relevant onboarding actions are captured in the audit trail with user, action, timestamp and status details	Planned

UAT Exit Criteria

- All critical business scenarios executed.
- No Critical or High severity defects remain open.
- Business stakeholders approve the solution.
- Functional requirements validated.
- UAT Sign-off pending.

22. Requirements Traceability Matrix

Business Requirement	Functional Requirement	User Story	UAT
BR-01	FR-01, FR-02	US-01	UAT-01
BR-02	FR-04, FR-05	US-04	UAT-03
BR-03	FR-06, FR-07	US-05, US-06	UAT-04
BR-04	FR-07	US-06	UAT-04
BR-05	FR-10	US-11	UAT-07
BR-06	FR-10, FR-11	US-11	UAT-07
BR-07	FR-12, FR-13	US-14	UAT-08
BR-08	BR-Rule-06	US-16	UAT-11

UAT Stakeholders

Stakeholder	Responsibility
Product Manager	Validate business objectives
Business Analyst	Prepare test cases, coordinate UAT, support issue resolution
Operations Team	Validate onboarding workflow
Compliance Team	Verify regulatory scenarios
QA Team	Support defect validation
Engineering Team	Fix identified defects
Business Sponsor	Final business sign-off

Defect Severity Matrix

Severity	Description	Expected Resolution
Critical	Core business function unavailable	Immediate
High	Major functionality impacted	Before release

Medium	Partial functionality affected	Planned fix
Low	Cosmetic or minor usability issue	Future release

UAT Sign-off

Role	Sign-off Status
Product Manager	Not Executed
Business Analyst	Not Executed
Operations Manager	Not Executed
Compliance Manager	Not Executed

Sections in Phase 8

23. Data Analytics Overview

The Healthcare Digital Platform captures onboarding lifecycle data to monitor operational efficiency, provider activation, SLA compliance, and exception handling. SQL queries transform transactional data into actionable KPIs, which are visualized through an executive Tableau dashboard.

24. SQL Analytics

SQL Analysis	Business Purpose
Provider Approval Rate	Monitor onboarding success
Average Verification Time	Measure operational efficiency
Pending Verification Queue	Identify bottlenecks
Rejection Reasons	Analyze onboarding failures
SLA Compliance	Track processing timelines
Provider Activation Rate	Measure onboarding effectiveness

25. KPI Framework

KPI	Formula	Business Value
Total Applications	Count(Applications)	Overall workload
Approval Rate	Approved / Total	Onboarding efficiency
Rejection Rate	Rejected / Total	Process quality
Pending Queue	Pending Count	Operational backlog
Avg Verification Time	Avg(Verification - Submission)	Team productivity
Avg Approval Time	Avg(Approval - Submission)	SLA monitoring
Provider Activation Rate	Active / Approved	Business growth

26. Dashboard Overview

- Executive KPI Cards
- Approval Trend
- Status Distribution
- Provider Performance
- SLA Compliance
- Operational Bottlenecks

27. Business Insights

Insight 1

Pending applications increase whenever verification turnaround exceeds SLA.

Recommendation:

Introduce automated reminders and escalation workflows.

Insight 2

Document mismatch is the primary rejection reason.

Recommendation:
Improve document validation during registration.

Insight 3

Automated verification reduces manual workload.

Recommendation:
Expand API integrations for additional verification checks.

Insight 4

Real-time dashboards improve operational visibility.

Recommendation:
Implement role-based dashboards for Operations and Compliance teams.

Decision Matrix.

Observation	Recommendation	Expected Business Impact
High Pending Queue	Auto-escalation	Faster approvals
Frequent Document Mismatch	Front-end validation	Lower rejection rate
SLA Breaches	Priority routing	Better compliance
Payment Failures	Retry workflow	Improved customer experience

Phase 9 – Governance, Compliance & Risk Management

28. Governance & Compliance

Objective

Ensure the provider onboarding process complies with regulatory requirements, protects sensitive data, and maintains complete auditability throughout the onboarding lifecycle.

Regulatory Compliance

Regulation	Application in Project
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DPDP Act 2023	Protect provider and patient personal information
Healthcare Data Privacy	Restrict unauthorized access to sensitive information
Internal Audit Policy	Maintain complete onboarding audit logs
Information Security Policy	Enforce secure handling of provider credentials

Role-Based Access Control

Role	Access Level
Provider	Manage own profile and documents
Operations Team	Review onboarding applications
Compliance Team	Verify credentials and approve/reject exceptions
Support Team	Manage customer issues and escalations
Product Team	View dashboards and reports
System Administrator	Full platform access

Audit & Logging

The system maintains a complete audit trail for every onboarding transaction, including:

- Provider registration
- Document submission
- Verification status changes
- Approval/Rejection actions
- Payment status updates
- Notification history
- User activity logs

29. Assumptions, Dependencies & Constraints

Assumptions

- Providers submit accurate registration details.
- External verification services are available.
- Required documents are uploaded during registration.
- Users have internet connectivity.

Dependencies

- National Medical Council Verification API
- Payment Gateway
- Notification Service
- Internal Operations Team
- Compliance Team

Constraints

- Third-party API availability
- Regulatory compliance requirements
- Network latency
- Data privacy policies
- SLA commitments

30. Risk Register

Risk	Probability	Impact	Mitigation
NMC API downtime	Medium	High	Retry mechanism and manual verification
Incorrect provider information	Medium	High	Mandatory validation and compliance review
Payment failures	Medium	Medium	Retry workflow and support escalation
SLA breaches	Medium	High	Automated alerts and priority queues
Unauthorized data access	Low	High	RBAC and encryption
Notification delivery failure	Low	Medium	Retry through alternate channels

31. Assumptions & Decision Log

Decision	Reason
Automated verification introduced	Reduce manual effort
Dashboard implemented	Improve operational visibility
Rule-based approval	Standardize decision making
Payment module included	Support end-to-end consultation lifecycle
Notification automation	Improve stakeholder communication

32. Security Considerations

Area	Control
Authentication	Secure user login
Authorization	Role-Based Access Control (RBAC)
Data Encryption	Encrypt sensitive data in transit and at rest
Audit Logging	Record all business-critical actions
API Security	Token-based authentication
Password Security	Encrypted password storage

Governance Checklist

Item	Status
Data Privacy	Addressed
Audit Logging	Addressed
Access Control	Addressed
Regulatory Compliance	Addressed
Risk Management	Addressed
Security Controls	Addressed

Phase 10 – Project Closure & Business Outcomes

33. Business Impact Assessment

Expected Business Benefits

Business Area	Expected Outcome
Provider Onboarding	Reduced onboarding turnaround time through workflow automation
Operational Efficiency	Reduced manual verification effort and standardized processes
Regulatory Compliance	Improved compliance through automated verification and audit trails
Provider Experience	Faster onboarding and real-time application status updates
Customer Support	Reduced support effort through automated notifications and issue tracking
Executive Reporting	Improved operational visibility through dashboards and KPIs

Key Performance Indicators (KPIs)

KPI	Baseline	Target
Average Onboarding Time	5 Business Days	< 24 Hours
Provider Approval Rate	50% <i>(sample dataset)</i>	> 80%
Manual Verification Effort	High	Reduced through automation
SLA Compliance	70%	> 95%
Dashboard Availability	Not Available	Real-Time Visibility

34. Project Deliverables

Deliverable	Status
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Business Requirements Document (BRD)	Completed
Functional Requirements	Completed
Non-Functional Requirements	Completed
Business Rules	Completed
Functional Architecture	Completed
ERD & Data Dictionary	Completed
API Contracts	Completed
Agile Backlog (16 User Stories)	Completed
Gherkin Acceptance Criteria	Completed
UAT Test Pack	Prepared
SQL Analytics	Completed
Tableau Dashboard	Completed
Governance & Compliance	Completed

35. Lessons Learned

Key Learnings

- Early stakeholder alignment reduces requirement changes.
- Automating verification improves operational efficiency.
- Standardized business rules improve process consistency.
- API integrations reduce manual dependency and turnaround time.
- Executive dashboards enable faster, data-driven decision-making.
- Well-defined UAT scenarios improve release quality and business confidence.

36. Future Enhancements

Enhancement	Business Value
AI-assisted document validation	Reduce manual document review
OCR-based certificate extraction	Faster provider onboarding
Multi-registry verification	Broader regulatory coverage

Mobile provider onboarding	Improved accessibility
Predictive SLA risk alerts	Proactive operational management
Advanced operational dashboards	Deeper business insights

37. Appendix

Supporting Assets

- Healthcare Digital Platform PPT (Asset 1)
- Business Analysis Documentation (Asset 2)
- SQL Database Schema (schema.sql)
- SQL Analytics Queries (analytics_queries.sql)
- Sample Dataset
- Tableau Dashboard
- ERD Diagram
- AS-IS Swimlane Diagram
- TO-BE Swimlane Diagram
- API Specifications

Tools Used

Category	Tool
Documentation	Microsoft Word
Presentation	Microsoft PowerPoint
Database	PostgreSQL
SQL	PostgreSQL SQL
Dashboard	Tableau Public

Document Summary

Project Name: Healthcare Digital Platform

Domain: HealthTech | Provider Lifecycle | Patient Care | Digital Healthcare Operations

Role Demonstrated: Business Analyst

Methodology: Agile

Key Capabilities Demonstrated:

- Business Analysis
- Requirements Engineering
- Process Mapping
- Functional Design
- API Analysis
- Data Modeling
- SQL Analytics
- Dashboard Design
- User Story Writing
- UAT Planning
- Governance & Compliance
- Stakeholder Management